INVERSE FUNCTIONS

One-to-one, or no inverse

Different inputs must always produce different outputs.

f has an inverse $\iff$ f is one-to-one

Algebraic test: assume $f(a) = f(b)$ and try to prove $a = b$.

if $f(a) = f(b)$, then $a = b$

● ONE-TO-ONE

$f(x) = 3x + 7$

If $3a + 7 = 3b + 7$, then $3a = 3b$, so $a = b$.

Strictly monotonic functions

Only increasing or only decreasing, no flat stretches: they never produce the same output twice.

● NOT ONE-TO-ONE

$f(x) = x^2$

If $a^2 = b^2$, then $a = \pm b$, which does not imply $a = b$.

A function that turns around

Multiple inputs share an output; the reversal is ambiguous.

THE BOUNDARY Reversal needs each output to point to a unique input. The horizontal line test reads the same condition off the graph.